Here's the fix and code for all four tasks.

Task 1: Fix the bug in `trap_enemies`

The distance is squared (`distance_squared_to`), but the threshold `24 * 2` is 48 rather than 24². So the bubble only traps within about 7 pixels. Change it to:

if bubble.pos.distance_squared_to(enemy.center) < 24 ** 2:

Task 2: `bubble_tint(bubble)`

Empty bubbles keep the default colour. Bubbles with a trapped enemy shift from green to yellow to red as `bubble.life` runs out (it starts at 8.0), and flash white in the last 1.5 seconds.

```
def bubble_tint(bubble):
    """Return an (r, g, b) colour for a bubble, or None for the default."""
    if not bubble.enemy:
        return None
    frac = max(0.0, bubble.life / BUBBLE_LIFE)   # 1.0 = fresh, 0.0 = about to escape
    if bubble.life < 1.5 and int(bubble.life * 8) % 2 == 0:
        return (255, 255, 255)                   # warning flash
    if frac > 0.5:                               # green -> yellow
        t = (1 - frac) * 2
        return (int(80 + 175 * t), 255, 80)
    t = frac * 2                                 # yellow -> red
    return (255, int(70 + 185 * t), 70)
```

Task 3: `on_fruit_collected(fruit)`

A simple version is a counter that shows in the HUD. Use a module-level variable, since the function is outside `Game`:

```
fruits_collected = 0


def on_fruit_collected(fruit):
    """Called when the player picks up a fruit; add a sound, sparkle, or bonus effect here."""
    global fruits_collected
    fruits_collected += 1
```

Then show it in the HUD by changing the `hud = ...` line in `draw`:

```
hud = self.font.render(
    f"Score {self.score}  Lives {self.lives}  Level {self.level}  Fruit {fruits_collected}  R = reset",
    True, (240, 240, 240))
```

To reset the counter on `R`, add this at the top of `Game.reset`:

```
global fruits_collected
fruits_collected = 0
```

Task 4: `bonus_life_threshold()`

```
def bonus_life_threshold():
    """Return a score value at which the player earns an extra life, or None to disable bonus lives."""
    return 5000
```

For your "after" video, a smaller value like `1000` makes the bonus life easier to show in 10 seconds. You can change it back to `5000` afterwards.

Testing for your videos

- **Task 1:** Blow a bubble so it just grazes an enemy. It should now trap it, where before it flew past.
- **Task 2:** Trap an enemy and watch the bubble go green, yellow, red, then flash before the enemy escapes.
- **Task 3:** Pop a trapped bubble from above and grab the fruit. The HUD counter should go up.
- **Task 4:** Score enough to cross the threshold and watch `Lives` go up by 1.

Since the lab wants you to work iteratively with an LLM, it's worth asking me follow-ups too, like adding a sparkle effect for Task 3.

Link to chat : https://claude.ai/share/07147239-55ad-4ae1-bc60-219f58682ff8